

YARE-GAN: Yet Another Resting State EEG-GAN

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Abstract

Resting-state EEG offers a non-invasive view of spontaneous brain activity, yet the extraction of meaningful patterns is often constrained by limited availability of high-quality data, and heavy reliance on manually engineered EEG features. Generative Adversarial Networks (GANs) offer not only a means to synthesize and augment neural signals, but also a promising way for learning meaningful representations directly from raw data, a dual capability that remains largely unexplored in EEG research.

In this study, we introduce a scalable GAN-based framework for resting-state EEG that serves this dual role: 1) **synthesis** and 2) **unsupervised feature extraction**. The generated time series closely replicate key statistical and spectral properties of real EEG, as validated through both **visual and quantitative evaluations**. Importantly, we demonstrate that the model's learned representations can be repurposed for a **downstream gender classification task**, achieving higher out-of-sample accuracy than models trained directly on EEG signals and performing comparably to recent EEG foundation models, while using significantly less data and computational resources.

These findings highlight the potential of generative models to serve as both neural signal generators and unsupervised feature extractors, paving the way for more data-efficient,

architecture-driven approaches to EEG analysis with reduced reliance on manual feature engineering. The implementation code for this study is available at: <https://github.com/Yeganehfrh/YARE-GAN>

Keywords: Generative adversarial networks (GANs); EEG; Unsupervised Representation Learning

Introduction

Generative AI, including techniques such as Generative Adversarial Networks (Goodfellow et al., 2014), Variational Autoencoders (Kingma, 2013), and diffusion models (Ho et al., 2020), is being increasingly recognized as a powerful tool in neuroscience. It has found various applications ranging from synthesizing neural data to also learning reusable representations of neural signals in an unsupervised manner.

One of the challenges in neuroscience is the limited availability of high-quality neural data, as collecting such data is often costly and time-consuming. Generative AI can help address this issue by creating realistic synthetic datasets that capture the statistical properties of real neural recordings. Additionally, in cases where labeled data is sparse or certain conditions are underrepresented, synthetic data can help balance datasets, leading to more robust analyses and improved model performance (Murphy, 2022). The use of generative AI, particularly GANs, for data augmentation in neural datasets has been relatively well studied (Williams 2025). Prior research has demonstrated that augmenting datasets with generated neural signals can enhance classification accuracy in tasks such as motor imagery (Hartmann et al., 2018; Fahimi et al., 2020), emotion recognition (Pan & Zheng, 2021), and detection of epileptic seizure (Pascual et al., 2020).

Beyond data augmentation, generative AI also provides a way to learn meaningful feature representations in an unsupervised manner (Radford, 2015). By learning a latent space that effectively captures the underlying structure of neural data, GANs can generate interpolated

samples that preserve key statistical and physiological properties. These learned representations can be repurposed for supervised tasks, reducing the need for extensive and often labor-intensive feature engineering. This approach is particularly beneficial for neural signal processing, where preprocessing and manual feature extraction can significantly influence final results (Robbins et al., 2020; Botvinik-Nezer et al., 2020). This way, an unsupervised generative model can first be trained using publicly available, large-scale resting-state datasets and later fine-tuned on task-specific data, which is often more limited. In this way, generative AI not only mitigates the issue of limited case-specific samples through data augmentation but also tackles this issue by offering transferable, reusable representation of the data that can later be fine-tuned for a downstream, case-specific supervised task.

While recent work has introduced novel architectures like criss-cross transformers for unsupervised EEG representation learning (Wang et al., 2024), the use of generative adversarial networks (GANs) for this purpose remains relatively underexplored, with some studies investigating their potential (Liang et al., 2021). This may be due to the fact that GANs in neuroscience have primarily been applied for data augmentation, often on small, domain-specific datasets, and mainly focused on generating extracted features rather than raw time-series data. As a result, GANs have not been widely utilized as direct neural signal processors or feature extractors.

In this study, we aim to extend the use of GANs beyond data augmentation by incorporating larger-scale EEG recordings—particularly resting-state, task-free data—to train a GAN model for neural data generation. We further explore the utility of the learned representations from its intermediate layers of its discriminator for downstream classification tasks, demonstrating their potential for improving neural signal analysis.

Methods

Data

We used the MPI-LEMON dataset (Babayan et al., 2019), a publicly available resource for investigating mind–body–emotion interactions. This dataset includes resting-state EEG (rs-EEG) recordings from 216 participants, acquired with 61 EEG channels during both eye-closed and eye-open conditions. Data from 13 participants were excluded due to missing event information, inconsistent sampling rates, mismatched header files, or poor signal quality (Babayan et al., 2019). Each participant contributed a 16-minute EEG recording (8 minutes per each condition), resulting in a total of 54.13 hours of rs-EEG data.

Preprocessing

To preserve the natural structure of the EEG data, we kept preprocessing to a minimum (Delorme, 2023). We first downsampled the EEG signals to 128 Hz. Then, following the steps outlined in (Défossez et al., 2023), we applied Baseline correction by subtracting the average value of the first 0.5 seconds from each channel. The data was then normalized using Scikit-Learn’s robust scaler, followed by standard scaling. To mitigate the impact of extreme outliers, values exceeding 20 standard deviations were clamped. Data clamping is found to be as effective as more complex artifact rejection methods such as AutoReject (Défossez et al., 2023).

EEG signals were then high-pass filtered at 0.5 Hz to eliminate low-frequency drift. To suppress line noise at 50 Hz, we then applied the Denoising Source Separation (DSS) method (Cheveigne, 2010). The cleaned signals were subsequently segmented into around 5-seconds epochs (512 time points each). In this study, we included eight EEG channels (F1, F2, C1, C2, P1, P2, O1, and O2). Selecting 8 out of the 61 EEG channels was primarily for practical reasons, since this subset allowed for faster experimentation and model development. However, the proposed model is designed to be scalable. Specifically, the capacity of each layer, such as the number of filters in the convolutional layers, is proportional to the number of input channels.

This allows the model to be adapted to accommodate and process additional EEG channels as needed. In the supplementary materials we compared the model's performance with additional 16 and 56 channels to demonstrate the scalability of the results.

Architecture Details

We implemented a Wasserstein Generative Adversarial Network (Arjovsky et al., 2017) with **Gradient Penalty (WGAN-GP)** (Gulrajani et al., 2017) to generate multi-channel EEG data. The model consists of two core components: A generator, responsible for producing synthetic EEG signals, and a discriminator—referred to as a "Critic" in the WGAN framework—which evaluates whether the generated data is real or synthetic.

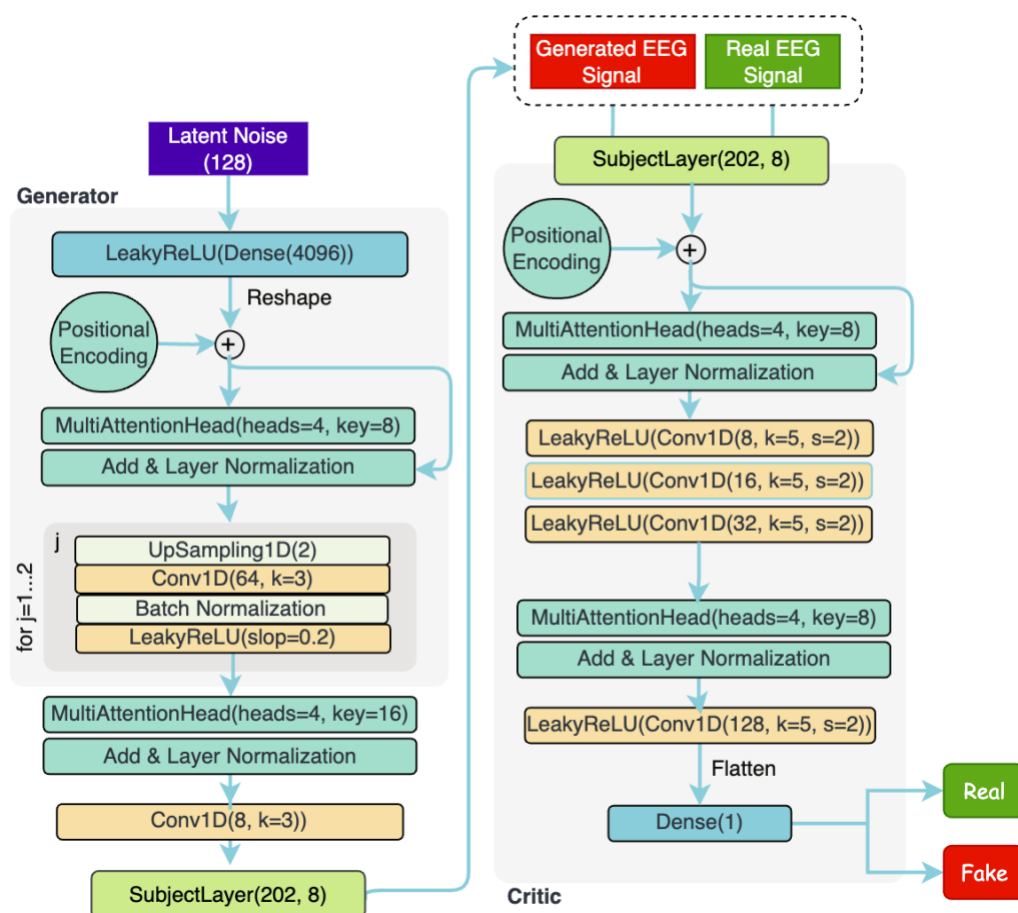


Figure 1: Architecture of the proposed model. The general structure of this model combines a DCGAN-inspired architecture with self-attention layers and positional encoding to better model temporal dependencies in EEG time-series data. The capacity of each layer (such as the number of filters in the convolutional layers) is scalable according to the number of EEG channels.

Our architecture follows the Deep Convolutional GAN (DCGAN) (Radford, 2015) framework, incorporating a stack of convolutional layers with upsampling layers in the generator and strided convolutional layers in the Critic (Figure 1). For upsampling, we used linear interpolation, as it has been empirically shown to produce significantly fewer high-frequency artifacts compared to nearest-neighbor upsampling (Hartmann et al., 2018).

To improve the model’s ability to capture long-range dependencies in EEG signals, we integrated a one-dimensional self-attention mechanism into both the generator and the Critic. This self-attention layer includes: (1) A multi-head attention mechanism applied along the time dimension. (2) a post layer normalization, (3) and a learnable positional embedding layer, similar to how they are used in Transformers. Instead of using fixed sinusoidal embeddings (as in Vaswani, 2017), we use trainable positional embeddings for each time step, which are added to the input tensor before feeding the sequence into the multi-head attention block. Our ablation experiment showed that this self-attention layer is essential in model performance in generating realistic and diverse EEG data. All weights in the convolutional and dense layers were initialized using a random normal distribution with a mean of 0 and a standard deviation of 0.02. This initialization has been shown to stabilize GAN training and prevent divergence (Radford, 2015).

To account for individual variability across participants and enhance diversity in the generated EEG signals, we incorporated a participant-specific transformation layer. Inspired by Défossez et al., 2023; this layer, which is applied after the generator produces synthetic data, learns a

participant-specific transformation matrix $M \in R^{D1 \times D1}$ for each participant $s \in [S]$. The subject layer essentially acts as a spatial filter per subject that makes each sample's channels linearly mixed according to their subject-specific transformation. Since we can assume that each subject might represent EEG signals slightly differently across channels, this transformation gives the model a chance to adapt by learning a unique projection per subject.

This modification was critical in preventing generator mode collapse, a scenario where the generator produces only a limited set of examples to deceive the critic. By leveraging this layer, the generator was able to produce more diverse outputs.

Training

For the training of this model, we used the Wasserstein loss function with gradient penalty.

$$L = E_{z \sim p(z)}[D(G(z))] - E_{x \sim p_r}[D(x)] + \lambda E_{\hat{x} \sim p_{\hat{x}}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2]$$

The first term estimates the distance between the real data distribution and the generated data's distribution using the Wasserstein distance (also known as the Earth Mover's Distance).

The second term, known as gradient penalty, ensures that the Critic function remains close to 1-Lipschitz by penalizing deviations of the gradient norm from one. $P\hat{x}$ denotes the distribution of points obtained by sampling uniformly along the line segments between real and generated samples. λ is the gradient penalty coefficient. We set this coefficient to 10 based on the original WGAN-GP paper, but we also tested alternative values and retained the one that yielded the most stable training.

The generator loss is defined as:

$$L_G = E_{z \sim p(z)}[D(G(z))]$$

reflecting the generator's objective to produce samples that maximize $D(G(z))$ (or equivalently, minimize $-D(G(z))$), thereby reducing the estimated Wasserstein distance between the generated and the true data distributions.

Both the generator and critic were optimized using the Adam optimizer, with $\beta_1 = 0.5$, $\beta_2 = 0.9$, and with an initial learning rate of 0.00094 and a batch size of 128. The learning rate was reduced progressively using an exponential decay schedule with a decay step of 100,000 and a decay rate of 0.90. This gradual reduction in the learning rate was crucial to preventing training instabilities, such as model blow-ups or divergence. Training parameters (e.g. batch size, learning rates) were tuned empirically to ensure stable GAN training and avoid divergence or collapse. We chose a relatively large batch size (128), as our hardware allowed for it. This improved gradient estimates but required lowering the learning rate to maintain stability.

Latent variables z for the generator was sampled from a normal distribution with the mean and standard deviation of the real data. To ensure the Critic remains well-trained and provides meaningful feedback to the Generator, the Critic was updated twice per each Generator update. Specifically, in each training iteration, two full forward and backward passes were performed on the Critic using both real and generated samples before updating the Generator once.

Data shuffling across batch dimensions was enabled in the Keras fit function, ensuring that each training iteration included segments from different participants. The model was trained for 1200 epochs (129,600 steps) on an NVIDIA V100 GPU with mixed precision using CUDA 12.7. The entire model development and training process was implemented using Keras (v3.5.0) with the PyTorch (v2.4.1) backend. The implementation code for this model is available at: <https://github.com/Yeganehfrh/YARE-GAN>

Evaluation Metrics

To assess the quality of the generated EEG data, we conducted both visual inspections and quantitative comparisons between real and generated signals. We first examined the generated signals in the time and frequency domains to check for realistic waveform structures and

spectral properties. Next, we quantified the similarity between real and synthetic data by extracting several statistical and spectral features, including:

Hjorth parameters (activity, mobility, complexity) to capture signal variance, frequency composition, and dynamic complexity (Hjorth, 1970).

Higher-order statistics, such as kurtosis and skewness, to analyze the distribution of EEG amplitudes.

Relative Power spectral features across standard EEG frequency bands (delta, theta, alpha, beta, gamma) to compare oscillatory activity.

Functional connectivity between EEG channels to ensure that the generated data maintained realistic inter-channel relationships. We computed pairwise cosine similarity between EEG channels as a proxy measure of functional connectivity.

To track how well the model learned the statistical properties of real EEG, we computed the **Fréchet Distance (FD)** between the feature distributions for each EEG channel throughout training, measured every 50 epochs. This allowed us to monitor the learning process, identify potential challenges in capturing specific channels' features.

Classification task: reusing the Critic's intermediate representations for a downstream task

To assess whether the representations learned in the intermediate layers of the GAN were useful for a new task, we repurposed the Critic for gender classification. Specifically, we extracted data representations from two intermediate convolutional layers of the Critic, and used it as inputs to two separate classifiers.

1. **Fully Connected Classifier** – A dense layer with 512 units and a GELU activation, followed by dropout (rate = 0.4), and a single-unit output layer with sigmoid activation.

2. **Convolutional Classifier** – A one-dimensional convolutional layer (64 filters, kernel size = 5) with GELU activation, followed by max pooling (pool size = 2). The output was flattened and passed through a dense layer with 128 units before the final single-unit sigmoid layer.

The Critic’s extracted features had a shape of (128, 32), which were fed directly into the convolutional classifier. For the fully connected classifier, these features were first flattened into a 4096-dimensional activation vector.

For this classification task, we used a publicly available EEG dataset from OpenNeuro (Kekecs et al., 2023), entirely independent of GAN training. The dataset comprised resting-state EEG from 52 participants (39 female, mean age 24.5), each with approximately 30 minutes of data. To achieve a balanced gender distribution, we resampled the data to include an equal number of male and female participants, resulting in a final subset of 26 participants. For train-test splitting, we used GroupShuffleSplit from Scikit-Learn to prevent data leakage by ensuring that EEG segments from the same participant did not appear in both the training and test sets (70% train, 30% test).

Models were trained with binary cross-entropy loss and the Adam optimizer (learning rate = 0.0001) for 1000 epochs, with a batch size of 256. Accuracy served as the primary evaluation metric.

As baselines, we trained the same two classifier architectures directly on minimally processed EEG data from the held-out dataset (preprocessing details described in the Methods). For the fully connected baseline, each EEG segment was flattened across the time (512) and channel (8) dimensions into a 4096-length vector. For the convolutional baseline, raw segments were fed directly into the convolutional layer.

For benchmarking, we compared against CBraMod (Wang et al., 2024), a recently proposed EEG foundation model that has achieved state-of-the-art accuracy on multiple downstream

tasks. We trained the same classifiers (as described above) on embeddings extracted from the CBraMod model, with no fine-tuning applied to either our model or CBraMod for this downstream task. To ensure a fair comparison, we preprocessed the held-out EEG dataset using the procedure as theirs. Specifically, we selected the same eight EEG channels as in our training set, downsampled the signals to 200 Hz, applied a bandpass filter between 0.3 Hz and 75 Hz, and removed line noise with a 50 Hz notch filter. Then we segmented the data into both four-second segments (800 time points; 6400 features) and two-second segments (400 time points; 3200 features), to account for the effect of segmentation length on feature dimensionality. For input to the convolutional classifier, we reshaped these feature vectors to match the original expected input shape of the model, resulting in dimensions of (200, 32) for four-second segments and (200, 16) for two-second segments, that corresponds to 200 time steps and a feature dimension of 8 channels multiplied by the segment size in seconds.

Results

Training Statistics

[Figure 2](#) shows the generator and Critic loss curves over 1200 training epochs. Throughout this process, we also continuously monitored the gradient penalty, the norm of the critic’s gradient, and the critic’s prediction of the real and fake data, to ensure stable learning and prevent divergence ([Figure 2](#)).

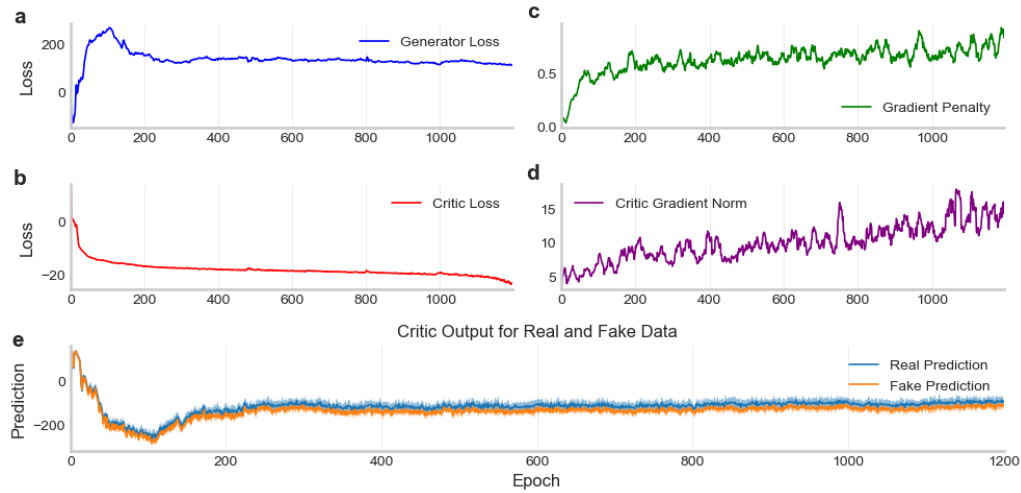


Figure 2: Training Statistics. Each graph shows key metrics tracked over 1200 training epochs: (a) Loss curves for the Generator (b) and Critic. (c) Gradient penalty (the L2 norm of the gradient of the Critic with respect to its input data) and (d) Critic gradient norm (the gradient magnitude of the Critic with respect to its weights). (e) The average Critic predictions for real and fake data at each epoch. The Critic consistently assigns higher values to real data, indicating that it continues to learn and has not been fully deceived by the Generator.

Result 1: The proposed architecture is successful in Generating multi-channels EEG data

Figure 3 presents a visual comparison between real and generated EEG signals, highlighting their average waveforms and standard deviations in both the time and frequency domains. Despite the model not being explicitly trained on frequency-domain data, the power spectral density (PSD) analysis shows that it successfully captured the spectral characteristics of different EEG channels.

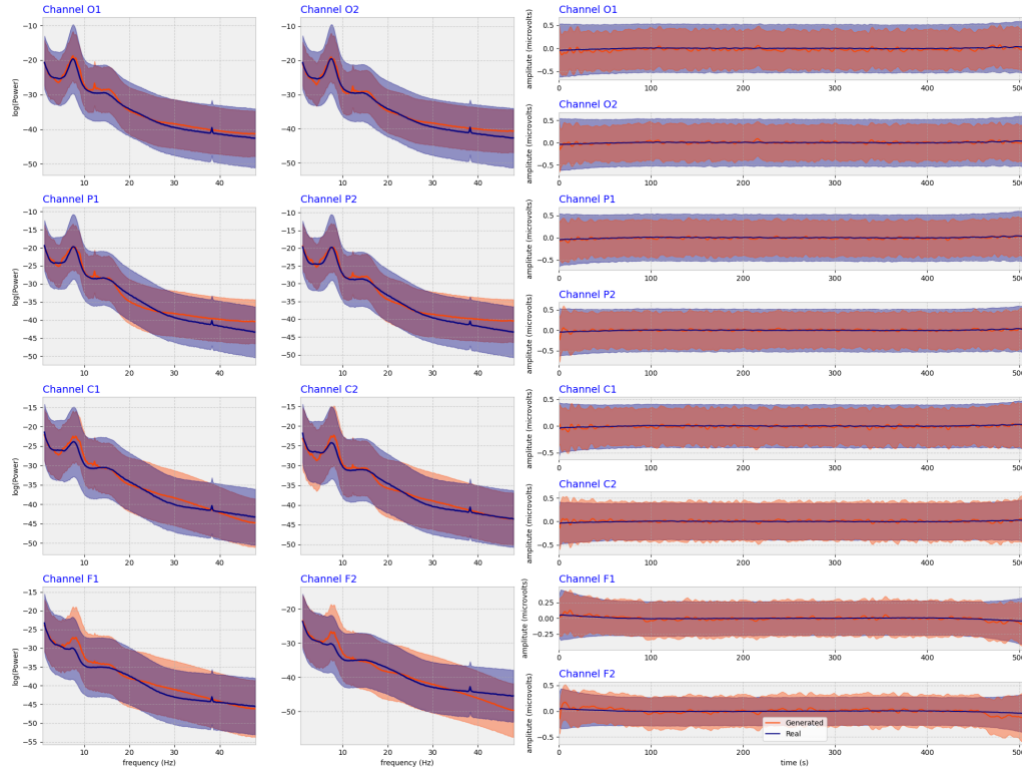


Figure 3: Visual inspection of generated and real data in the frequency and time domains across eight different EEG channels. The power spectral density (PSD) and time-series data are averaged over segments (batch dimension), with the shaded regions indicating the standard deviation. In all subplots, the blue line represents the real data, while the orange line shows the generated data.

In the time domain, signals remain within the same dynamic range demonstrating that the model effectively learned key temporal properties. Also, UMAP (Uniform Manifold Approximation and Projection) was used to assess the similarity between real and generated data in a lower-dimensional space. By reducing each EEG (512, 8) segment to 2 principal components, the UMAP plot reveals real and generated EEG data exhibit similar distributions in a lower-dimensional space [Figure 4](#).

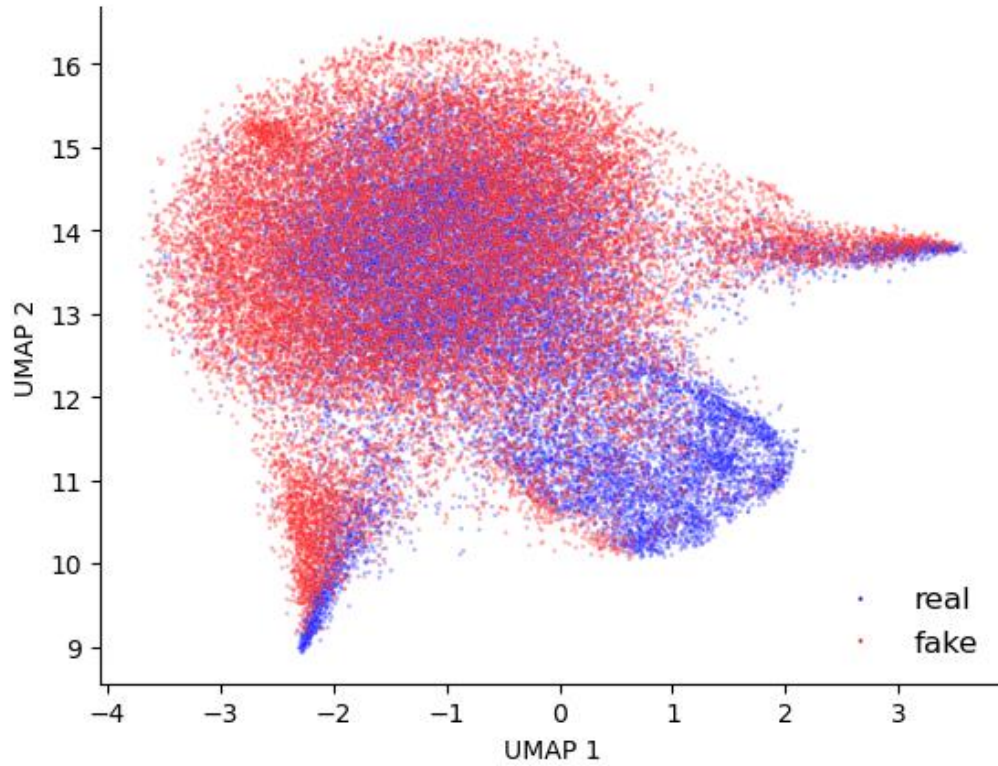


Figure 4: (a) UMAP visualization of real (blue) and generated (orange) EEG data in a two-dimensional space, showing the degree of overlap between the distributions. Each point represents an individual EEG segment.

To further strengthen the qualitative evaluation ("eye test"), [Figure 5](#) compares the functional connectivity of real and generated signals using cosine similarity. This adds a structure-sensitive perspective beyond time- and frequency-domain comparisons. The high degree of similarity in connectivity patterns suggests that the generative model successfully captures realistic inter-channel relationships in the EEG data.

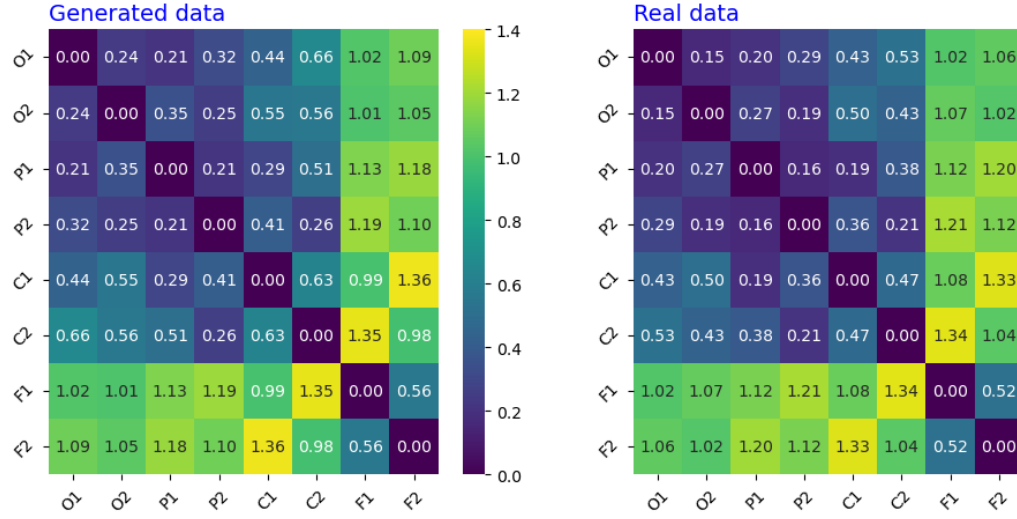


Figure 5: Cosine similarity matrices in generated and real data. The matrices show the similarity between signals from different EEG channels, demonstrating the model’s ability to reproduce realistic functional connectivity patterns

To quantify the similarity between real and generated EEG signals, we computed the **Fréchet Distance (FD)** on features extracted from both the time and frequency domains. Figure 6 illustrates how FD evolves during training across different electrode regions—frontal, central, parietal, and occipital.

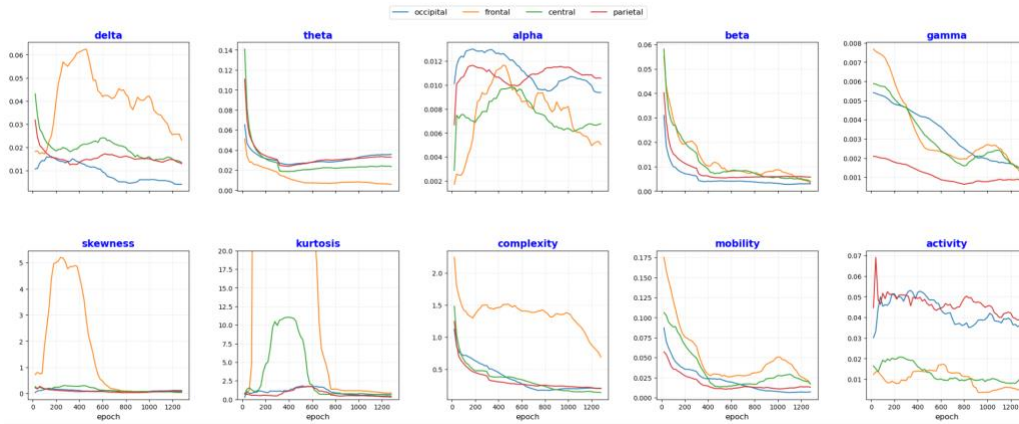


Figure 6: Fréchet Distance (FD) over training epochs. The upper row shows the FD between real and generated data for spectral features, measured as power within standard EEG frequency bands (delta, theta, alpha, beta, gamma) across different electrode regions. The lower row presents the FD evolution for manually extracted time-domain features, including Hjorth parameters (complexity, mobility, activity), skewness, and kurtosis. A decreasing FD over epochs indicates that the generated data increasingly resembles real EEG signals, though differences remain across frequency bands and electrode locations.

A breakdown of FD across EEG channels and frequency bands reveals that power in the delta, theta, and beta bands initially exhibits high FD values (up to $\sim 0.06 - 0.14$), followed by a steep decline within the first 200–300 training epochs indicating rapid early convergence. However, FD in the theta band remains relatively high (~ 0.03) throughout training, especially in parietal and occipital channels. This may reflect the bimodal and skewed distribution of theta power across samples (Figure 7), particularly in these posterior regions, which likely poses a greater challenge for the generator to model accurately.

Also, frontal electrodes, particularly in the delta band, consistently show slower and less stable convergence. This may be due to residual low-frequency noise—often introduced by artifacts such as sweating or eye blinks—that is more prevalent in frontal regions. Indeed, delta power distributions in F1 and F2 are notably broader, extending across higher delta power values, compared to posterior channels (Figure 7), supporting this interpretation.

Alpha band FD displays more variability, especially in frontal and central regions, but converges more smoothly in occipital and parietal areas, consistent with alpha’s typical dominance in posterior regions during resting-state EEG. Despite some fluctuations, FD values in the alpha band remain relatively low overall, peaking around ~ 0.012 .

Gamma band FD appears more jagged, but interestingly, it remains consistently low throughout training (~ 0.001 – 0.008). This apparent instability is perhaps due to the compressed y-axis scale rather than poor model performance, as the generator seems to approximate the gamma-band distribution well even from early epochs.

Spatial differences in generative fidelity are also evident across handcrafted time-domain features. As shown in the second row of Figure 6, FD for signal complexity remains high in frontal regions throughout training. This may reflect the inherent difficulty of modeling the more variable, higher-order dynamics associated with frontal EEG. However, it also underscores a limitation of using hand-engineered features to evaluate generative quality; statistical features like skewness and kurtosis, for instance, may not fully capture the nuanced structure that GAN is learning to mimic.

To complement the FD-based analyses, Figure 7 compares the empirical distributions of both real and generated data at two timepoints, early training (epoch 60) and final training (epoch 1200), for each feature across all channels. While this comparison confirms visible improvements over time, it also highlights the limitations of relying solely on traditional statistical descriptors. Moving forward, more informative evaluation metrics, such as those based on deep feature embeddings or similarity in model-learned latent spaces, may offer a more faithful reflection of generative quality in high-dimensional EEG data.

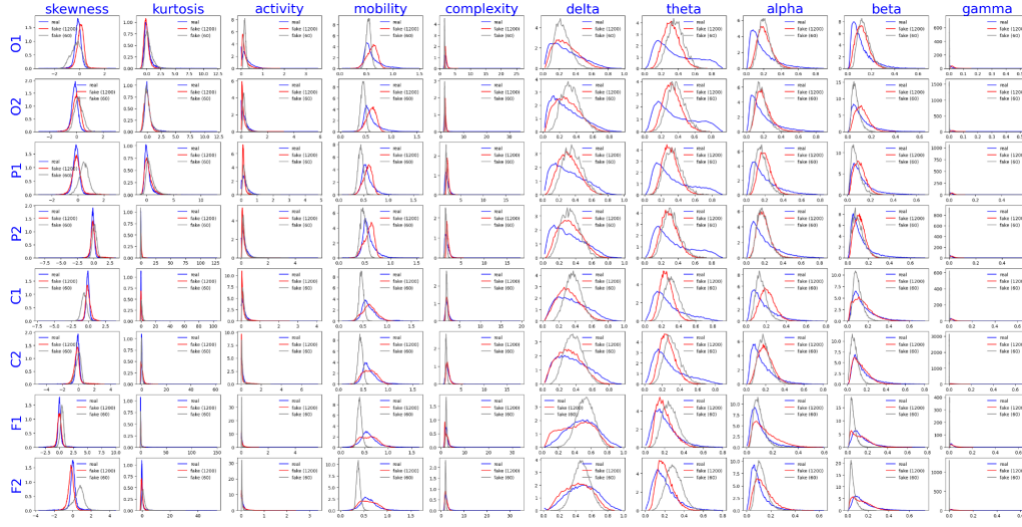


Figure 7: Comparison of real and generated EEG feature distributions across channels at early (epoch 60) and late (epoch 1200) training. Each subplot shows real (blue), early fake (gray), and late fake (red) distributions for handcrafted features (e.g., skewness, bandpower).

Result 2: The proposed Critic learns useful representation, making it reusable for a downstream task (gender classification)

To evaluate the transferability of the Critic’s learned representations, we tested its performance on a downstream gender classification task. As shown in Table 1, features extracted from our model’s Critic achieved a validation accuracy of 73% when used with a convolutional classifier, outperforming the same classifier trained on raw EEG data, as well as on features extracted from the CBraMod foundation model. In contrast, when paired with a simple fully connected classifier, the Critic’s representations yielded only 60% accuracy—barely above the baseline achieved on raw EEG and lower than CBraMod’s performance with the same classifier.

Despite this, our model required substantially fewer resources for pretraining: it was trained on less data and used significantly less computational power compared to CBraMod (see Table

1, final columns). These results highlight the potential of unsupervised representation learning via GAN-based architectures in EEG research, offering a scalable alternative to data-hungry foundation models and a promising direction for reducing dependence on manual feature engineering.

	Classifier Type	Validation Accuracy (%)	Feature Dimensionality	Pretraining Resources	Pretraining Data
Raw EEG Time Series	Convolutional Classifier	59.3 ± 0.49	(512, 8)	—	—
	FC Classifier	59.8 ± 0.70	4096	—	—
CbraMod (4s segments)	Convolutional Classifier	66.6 ± 0.48	(200, 32)	4× NVIDIA RTX A5000 GPUs, 5 days	~9,000 hours (~6.48B timepoints at 200 Hz)
	FC Classifier	66.8 ± 0.49	6400		
CbraMod (2s segments)	Convolutional Classifier	66.9 ± 0.58	(200, 16)		
	FC Classifier	64.5 ± 0.32	3200		

Ours	Convolutional Classifier	72.9 ± 0.53	(128, 32)	1× NVIDIA Tesla V100 16GB GPU, 1 day	~27 hours (~12.4M timepoints at 128 Hz)
	FC Classifier	60.2 ± 0.56	4096		

Table 1. Classification Performance Across Feature Sets and Classifiers. “*Validation accuracy*” (mean ± standard error) is reported for each feature extraction method and classifier type across training epochs. “*Raw EEG Time Series*” refers to the original input data. “*CbraMod*” and “*Ours*” represent feature sets extracted using the corresponding models; data segment lengths for *CbraMod* are indicated in parentheses. “*Convolutional Classifier*” and “*FC Classifier*” denote the downstream models applied to the extracted features. “*Feature dimensionality*” corresponds to the length of each input sample’s shape before classification. The top-performing result is bolded. Abbreviation: FC, Fully connected.

Discussion and Conclusion

In this study, we implemented a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) to generate multi-channel EEG data and evaluated the quality of the synthesized signals using both visual inspection and feature-based quantitative measures. Our results demonstrate that the model successfully learned key statistical, spectral, and inter-channel characteristics of EEG signals, as evidenced by decreasing Fréchet Distance (FD) over training epochs and visual resemblance between real and generated signals. Interestingly, the model achieved this without being explicitly trained on frequency-domain features or inter-channel connectivity—yet it still managed to preserve both spectral structure and spatial relationships between electrodes. The generator, however, struggles with replicating high-frequency components especially in the frontal region. This pattern aligns with the known

spectral bias of neural networks, particularly those using ReLU activations, which tend to learn low-frequency patterns before higher-frequency ones (Rahaman et al., 2019).

Compared to prior work, our approach offers several advancements. First, we propose a scalable architecture capable of generating multi-channel time series, in contrast to earlier studies that, while pioneering and foundational, focused on single-channel generation (e.g., Hartmann et al., 2018) or generating features instead of raw signals (e.g., Luo et al., 2020). Second, unlike prior studies that mostly employ GAN in task-specific BCI applications for data augmentation (e.g. Fahimi et al., 2020), our work uses task-free, resting-state EEG data, thereby increasing generalizability. Subsequently, these studies often relied on improved classification accuracy alone after data augmentation as a primary proxy for data quality, while broader metrics for signal diversity and realism are often overlooked (see Habashi et al., 2023, for a review). Furthermore, we leveraged a large dataset from hundreds of participants, and included a subject-specific transformation layer that accounts for individual variability—an important factor in neuroscience that is often overlooked in deep learning pipelines. This layer, which projects each subject’s EEG into a shared topographical space, helped stabilize training and enhanced the diversity of the generated signals.

Beyond signal generation, we evaluated the transferability of the learned features by using intermediate representations from the Critic for a downstream gender classification task. Without any fine-tuning, the Critic achieved 73% validation accuracy on out-of-sample data, when used with a convolutional classifier, outperforming a classifier trained directly on raw EEG. These results indicate that the Critic learned meaningful, generalizable EEG representations and highlights an often-overlooked aspect of GAN-based approaches: while most studies focus primarily on the generator’s output quality, the representational strength of the Critic is equally critical. Since the generator relies entirely on feedback from the Critic, a representationally weak or collapsed Critic limits the entire model’s performance. Our findings underscore the dual utility of our GAN architecture: the generator learns to produce

realistic EEG signals, while the Critic functions as a powerful unsupervised feature extractor—making it valuable for both generative and discriminative tasks in EEG research.

Also, the performance of our model is comparable to that of feature representations extracted from CBraMod, a recently proposed EEG foundation model, despite using significantly less training data and computational resources (Table 1). Although this comparison is limited (evaluating only one task), it nonetheless highlights the promise of GANs for both high-fidelity EEG generation and unsupervised feature extraction. As the field moves toward general-purpose foundation models and broader applications of generative AI, such dual-purpose GAN architectures may play an increasingly central role.

Despite the promising results, our approach has limitations. First, the training data was derived from a single dataset (LEMON) and limited to eight selected EEG channels, which may restrict the model’s ability to generalize to datasets with different recording conditions or additional EEG channels distributed across the scalp. Although we demonstrated that the current model can scale to include more EEG channels (see Supplementary Materials), further experimentation is still needed to achieve more stable training.

Building upon the current work, several directions can be pursued in future research; first, expanding the training data by incorporating larger, and more diverse EEG datasets, including task-related and clinical recordings, would improve the model’s generalizability. Additionally, enhancing the model’s ability to generate high-frequency EEG components represent a valuable goal. Further research could also examine broader applications of the learned representations, such as cognitive state decoding and neurological disorder detection. Finally, enhancing model interpretability through analysis of the GAN’s latent representations is essential. In particular, identifying which EEG channels contribute most to the learned features and classification outcomes, as well as examining attention weights in downstream classifiers, can shed light on the internal mechanisms of the model.

Overall, this study highlights the potential of GAN-based unsupervised learning to extract meaningful and generalizable EEG representations, paving the way for more data-efficient deep learning approaches in neuroscience.

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Supplementary Materials

Proposed Architecture's Scalability

To assess the scalability of the results, we compared the model's performance with additional 16 and 56 channels. In this setup, the architectural components such as the embedding dimension in positional encoding, the size of each self-attention head (query/key), and the number of feature maps in convolutional layers were scaled proportionally to the number of input channels. We trained these models for 150 epochs and compared their performance at the same training stage. Table (S.1) summarizes this comparison:

Number of Channels	8	16	56
Total Trainable Parameters	636,721	872,009	8,434,385
Training speed (Milliseconds per step)	197	200	350
FD (Spectral Features; normalized by number of channels)	0.163	0.173	0.965
FD (Hjorth parameters; normalized by number of channels)	0.182	0.179	4.168

The results indicate that training time does not scale linearly with the number of channels, an encouraging sign for scalability. The following graphs, ([Figure S.1](#)) compare the real and generated signals in terms of their power spectral density and connectivity matrix when 16 and 56 channels used, after 150 epochs of training.

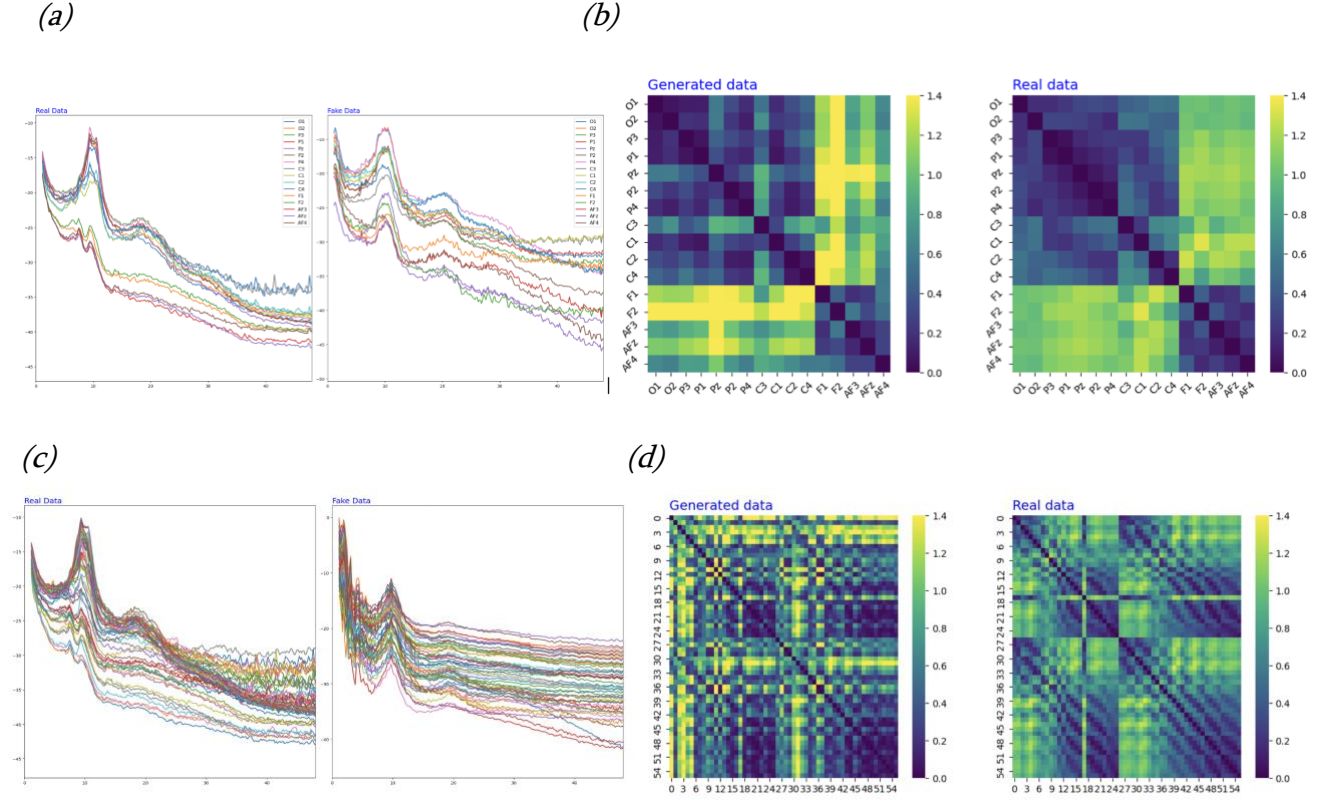


Figure S.1: Comparison between real and generated signals when using 16 and 56 EEG channels. The graphs show the results after 150 epochs of training. (a) power spectral density (PSD) of real and fake signals when using 16 EEG channels (b) Cosine similarity in real and fake when using 16 EEG channels. (c) PSD comparison when using 56 EEG channels (d) Cosine similarity comparison when using 56 EEG channels.